

Softplus#

<https://pytorch.org/docs/stable/generated/torch.nn.Softplus.html>

Description:

Applies the Softplus function element-wise. SoftPlus is a smooth approximation to the ReLU function and can be used to constrain the output of a machine to always be positive. For numerical stability the implementation reverts to the linear function when $\text{input} \times \beta > \text{threshold}$. β (float) – the β value for the Softplus formulation. Default: 1
threshold (float) – values above this revert to a linear function. Default: 20

Function Signature

```
class torch.nn.Softplus(beta=1.0, threshold=20.0)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.Softplus()  
>>> input = torch.randn(2)  
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies the Softplus function element-wise.

SoftPlus is a smooth approximation to the ReLU function and can be used to constrain the output of a machine to always be positive.

For numerical stability the implementation reverts to the linear function when $\text{input} \times \beta > \text{threshold}$.

`beta` (float) – the β value for the Softplus formulation. Default: 1

`threshold` (float) – values above this revert to a linear function. Default: 20

Input: $(*)^{(*)} (*)$, where $***$ means any number of dimensions.

Output: $(*)^{(*)} (*)$, same shape as the input.

Return the extra representation of the module.